Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/22/2005
PI Name:	Hunter, Gary Ph.D.		
Project Title:	Microsystems-Based Hydrazine Detection System for International Space Station and EVA Applications		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-01
Start Date:	07/15/2002	**End Date:**	07/15/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hong , Todd (Munez Engineering Inc./NASA Johnson Space Center) Young, Rebecca (NASA Kennedy Space Center) Makel, Darby (Makel Engineering, Inc.) Liu, C. (Case Western Reserve University)		
Grant/Contract No.:			
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Performance Goal Text:			

Task Description:	The detection of hydrazine at low concentrations (down to 10 ppb) is important for International Space Station Environmental Monitoring and EVA (Extra Vehicular Activity) applications. Low concentration of hydrazine can cause serious health problems for astronauts who can be exposed to hydrazine while on EVA mission outside of the International Space Station. Early detection of hydrazine is required both during an EVA, in the airlock, and as the airlock is opened into the Station. These applications require a potential hydrazine sensor to provide real-time response in vacuum, during pressurization, and in the presence of interfering gases. Present technology is not adequate for use in this Space Station application. Deficiencies include: 1. Inability to operate in the ISS environments; 2. Existing systems are too large for use where needed e.g. on the space suit; 3. Susceptibility to false alarm due, in part, to the presence of interfering gases; 4. Sensor has long response time or is hard to read; 5. Monitoring device does not interface readily to electronics to provide early warning signals. This task seeks to develop a microfabricated hydrazine sensor system to detect hydrazine for Space Station Environmental Monitoring and EVA applications. This sensor system will be developed using Schottky diode technology that has been shown to be effective in similar operational environments esp. leak detection on the Space Shuttle. A Schottky diode is a semiconductor-metal contact with possibly a thin interface layer in-between the metal and the semiconductor. Major challenges associated with this project are the very low concentrations measurements necessary and the need to measure in the presence of interfering gases. However, Schottky diode gas sensors have exponential responses, do not require oxygen for their operation, and are thus very well suited to this application. Various approaches will be investigated this project including the use of silicon carbide (SiC) as the semiconductor as well as various metal contacts and interface layers. Even if a sensor has short-term response the hydrazine, that does not mean that the sensor has long-term sensitivity and stability in a given environment. Thus, along with new approaches to hydrazine detection, long-term stability studies are necessary. Since interfering gas are present which could make determination of the exact hydrazine concentration difficult for a single sensor, the hydrazine sensor will be placed in an array. Leveraging other sensor and electronics development will be done as appropriate to produce a hydrazine sensor system to meet a significant Space Station technology need which presently cannot be addressed. This task brings together a highly qualified team which has a high degree of knowledge of the application combined with considerable experience in chemical sensor development, characterization, and application. The result of this project will not be flight hardware but prototype technology which performs in environments where other sensor technology is inadequate. This technology would be designed with flight consideration in mind and could later be qualified for flight use but that is not part of this present project. The purpose of this work is to improve astronaut health and safety by providing low concentration hydrazine detection with the following properties as previously determined by JSC personnel: 25 ppb minimum measurement of Hydrazine 5 minute response time 5 psi pressure operation Lack of corresponding response to ammonia. Improved monitoring of toxic materials such as hydrazine is important for present applicaitons such as International Space Station EVAs. Contaminant monitoring will continue to be mandatory in future exploration systems where contamination of closed cycled vehicle systems could have significant effects on astronaut health in long-term missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The detection of low concentrations of toxic chemicals has a broad range of earth related applications. While this technology is being developed specifically for hydrazine detection, its base technology has application to a range of other toxic chemicals as well.
Task Progress:	The demonstration of a Pd alloy based silicon-based Schottky diode system to detect ppb levels of hydrazine has been accomplished. Issues such as response time can be addressed to a significant degree using software approaches. The major steps associated with further development of this sensor is to verify its stability and reproducibility in long term operation and after exposure to other gases such as ammonia. Further, the use of atomically flat SiC, rather than Si or standard SiC surfaces, will be employed to hopefully improve the response as well as durability of the sensor.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Presentation	Hunter, G. W.; Neudeck, P.N.; Xu, J.; McFadden E.R.; Lukco, D.; Artale, M.; P. Lampard, P.; Androjna, D.; Liu, C. C.; Makel, D.; and Ward, B. "Micro Chemical Sensors and Sensor Arrays: Sensor System Development and Potential Healthcare Applications" Cleveland Clinic Lung Summit 2005 (Invited Talk). Apr-2005